

UV-Atlas Latent Dynamics Towards Sampling-Agnostic Cloth Simulation

Final Presentation

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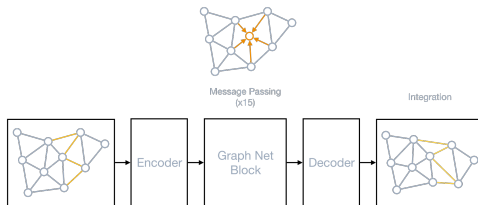
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Motivation



- Cloth simulation is central in computer graphics:
 - film and animation,
 - video games,
 - virtual try-on and interactive garment editing.
- Classical solvers can be realistic, but remain expensive.
- Learned simulators replace part of this cost with neural predictors.

Why Mesh-Based Learning Is Limited



MeshGraphNet architecture (Pfaff et al., 2021)

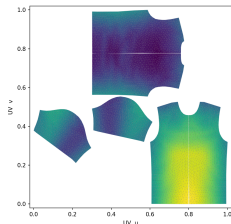
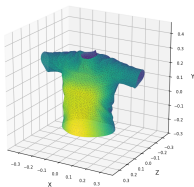
- MeshGraphNets and related methods operate directly on mesh vertices and edges.
- The recurrent computation scales with the number of vertices and edges.
- Local message passing needs many steps or a hierarchy to propagate long-range effects.

Pfaff et al. *Learning Mesh-Based Simulation with Graph Networks*. ICLR, 2021.

Grigorev et al. *HOOD: Hierarchical Graphs for Generalized Modelling of Clothing Dynamics*. CVPR, 2023.

Iparraguirre et al. *MeshGraphNet-Transformer*. arXiv, 2026.

Research Objective



UV atlas of a T-shirt mesh

- **Main idea:** Learn cloth dynamics in a fixed UV-atlas latent space, then decode positions at arbitrary compatible UV query points.
- **Desired behaviour:**
 - train on one cloth sampling,
 - evaluate on compatible lower-resolution samplings,
 - keep rollout behaviour stable when vertex count changes.

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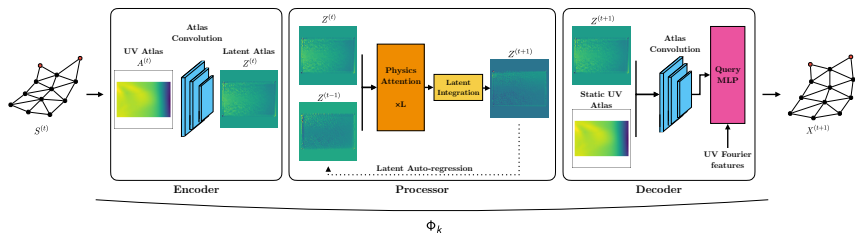
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Architecture Overview

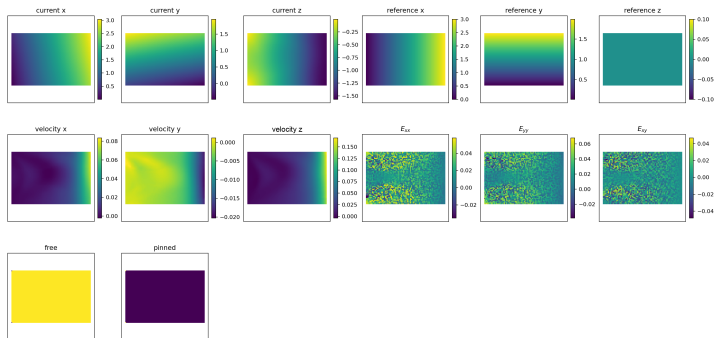


Overview of the proposed UV-atlas latent dynamics architecture.

- **Encoder E :** Atlas encoder converts the mesh state into a latent atlas map.
- **Processor P :** Latent processor combines local atlas convolutions with physics-attention blocks.
- **Decoder D :** Query decoder samples latent features at UV positions and predicts 3D vertex positions.

$$\Phi_k(S^{(t)}) = (D \circ P^k \circ E)(S^{(t)}) = X^{(t+k)}.$$

UV-Atlas Representation



Unnormalised input atlas channels from an ARCSim flag simulation.

- Vertex and face quantities are **rasterised** through UV coordinates.
- **Cross-atlas convolutions** preserve neighbourhoods across seams.

Li et al. *Cross-Atlas Convolution for Parameterization Invariant Learning on Textured Mesh Surface*. CVPR, 2019.
Santesteban, Otaduy, Casas. *SNUG: Self-Supervised Neural Dynamic Garments*. CVPR, 2022.

- The processor receives the current latent state and a latent velocity term:

$$Z^{(t)} - Z^{(t-1)},$$

and computes a latent acceleration $a^{(t)}$.

- The next state is integrated with a Verlet-style update:

$$Z^{(t+1)} = Z^{(t)} + \left(Z^{(t)} - Z^{(t-1)} \right) + a^{(t)}.$$

Training Objective

- **Semi-supervised learning:**

- latent loss,
- decoded position loss,
- physical loss.

$$\mathcal{L}^{(t)} = \mathcal{L}_{\text{latent}}^{(t)} + \mathcal{L}_{\text{position}}^{(t)} + \mathcal{L}_{\text{physics}}^{(t)}$$

- Physics-based minimisation on decoded meshes:
 - inspired by the optimisation problem from SNUG
- Limited-horizon prediction and weighted accumulation:

$$\mathcal{L} = \frac{\sum_{k=1}^n \gamma^k \mathcal{L}^{(t+k)}}{\sum_{k=1}^n \gamma^k}$$

Santesteban, Otaduy, Casas. *SNUG: Self-Supervised Neural Dynamic Garments*. CVPR, 2022.

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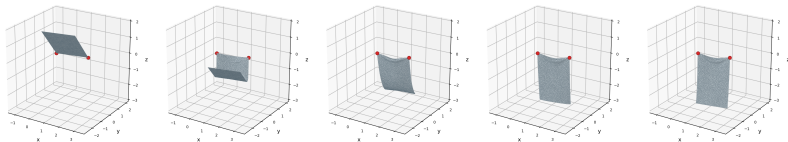
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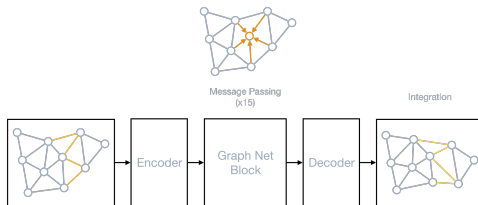
Dataset



Example ARCSim flag rollout, sampled every 20 frames.

- Rectangular flag, gravity only, two pinned vertices.
- 20 simulations of 100 steps each.
- Original training mesh: Poisson disk sampling with about 1.6k vertices.
- Tested on lower-resolution Poisson and uniform samplings.

Baselines and Metric

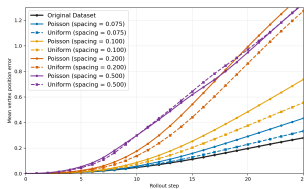


MeshGraphNet architecture (Pfaff et al., 2021)

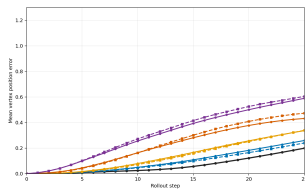
- Baselines:
 - MeshGraphNet (MGN),
 - MeshGraphNet-Transformer (MGN-T),
- Evaluation metric:

$$\text{MVE}^{(t)} = \frac{1}{|V|} \sum_{i \in V} \left\| \hat{x}_i^{(t)} - x_i^{(t)} \right\|_2.$$

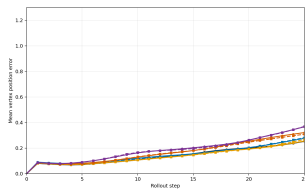
Mean Vertex Error Over Rollout



MGNet



MGNet-T



Ours

- Graph-based methods perform well on the original mesh but degrade with undersampling.
- The proposed method shows more consistent predictions across mesh samplings, indicating lower sensitivity to changes in vertex sampling.

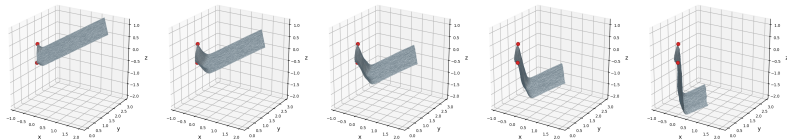
Sampling Robustness Summary

Metric	MGN	MGN-T	Ours
Mean MVE ↓	0.345	0.199	0.164
Std. ↓	0.169	0.089	0.020
CV ↓	0.491	0.446	0.120

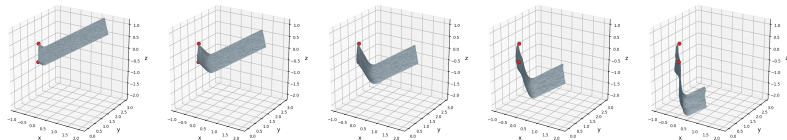
- Mean computed across all evaluated samplings and validation rollouts.
- Our method achieves the best mean error and the lowest variance across samplings.

Rollout at Low Sampling

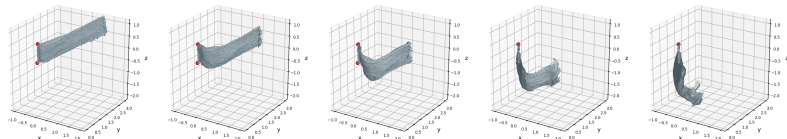
MGN



MGN-T

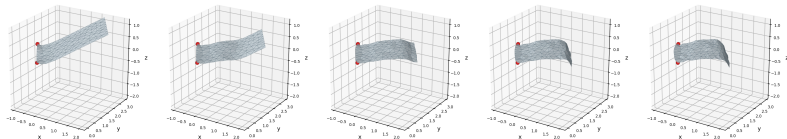


Ours

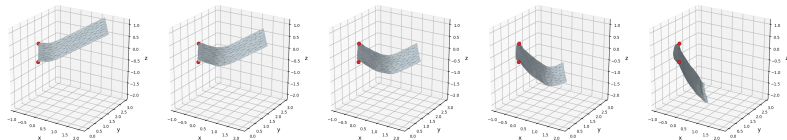


Rollout at Low Sampling

MGN



MGN-T



Ours

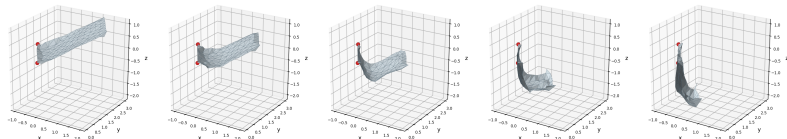
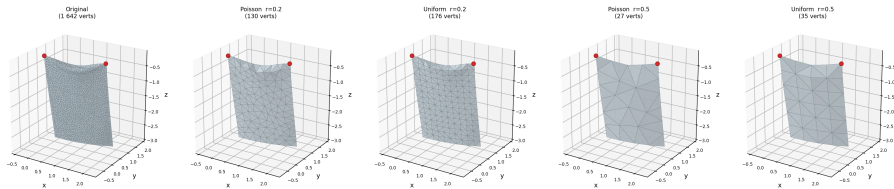


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Outcomes



Visualisation of different sampling levels and methods on the flag simulation

- These results support the central hypothesis: decoupling dynamics from the mesh topology makes the model robust to vertex count changes.
- On undersampled meshes, our model becomes more competitive and often achieves the best performance.
- On the original mesh, MGN-T still achieves the lowest error.

Limitations and Future Work

- Current evaluation is limited:
 - Evaluation uses only a gravity force,
 - Not compared against the latest baselines,
 - No atlas seams were used.
- No contacts, self-collisions, body interactions, or animated handle points.
- **Future directions:**
 - collision-aware inputs or losses,
 - animated constraints and external controls,
 - more diverse cloth motions, materials, and UV parametrisations.

Thank you for your attention!

References

- Pfaff et al. *Learning Mesh-Based Simulation with Graph Networks*. ICLR, 2021.
- Santesteban et al. *SNUG: Self-Supervised Neural Dynamic Garments*. CVPR, 2022.
- Grigorev et al. *HOOD: Hierarchical Graphs for Generalized Modelling of Clothing Dynamics*. CVPR, 2023.
- Liao et al. *SENC: Handling Self-Collision in Neural Cloth Simulation*. ECCV, 2024.
- Wu et al. *Transolver: A Fast Transformer Solver for PDEs on General Geometries*. ICML, 2024.
- Zhang et al. *ClothTransformer: Unified Latent-Space Transformers for Scalable Cloth Simulation*. arXiv, 2026.
- Iparraguirre et al. *MeshGraphNet-Transformer: Scalable Mesh-Based Learned Simulation for Solid Mechanics*. arXiv, 2026.